

SpaceCDF Learner's Workbook

*Worksheets and tool exercises for the 40-hour Concurrent
Design Facility intensive*

SpaceCDF

2026

How to use this workbook

This workbook is your hands-on companion through the 40-hour SpaceCDF intensive. There is one worksheet per session in the

Course Plan

. Each worksheet is single-page and lives next to the SpaceCDF tool tab where you'll do the work.

The workbook is paired with the

Facilitator's Book

, which carries the teaching content and worked solutions.

EXPECTED READING

Expected reading Each worksheet lists its expected reading at the top. Read it the night before — the worksheets are designed to be completed quickly when you arrive prepared.

The four templated tables that recur throughout this workbook (and which you will fill in repeatedly) are:

Template	When to use
Stakeholder matrix	Session 1.2
Trade-study matrix	Sessions 1.3, 3.4, 4.1
ConOps timing table	Session 1.4
SMART requirement check-list	Session 2.1
Functional decomposition table	Session 2.2
N ² matrix (blank)	Session 2.3
Equipment selection table	Sessions 4.1
Mass / power / link / Δv budget	Sessions 3.1 – 3.4
Risk register	Session 4.3
V&V matrix	Session 4.2
Cost-by-WBS	Session 4.4
Anomaly response form	Week 3
Self-reflection	End of every worksheet

Worksheet 1.1: Introduction to Space Mission Design

Name: _____ Date: _____

Part A: The 17 SE Processes -> SpaceCDF Mapping

For each NASA SE process, identify which SpaceCDF feature (tab, button, or panel) implements it. Mark “Manual” if it requires human judgment.

#	SE Process	SpaceCDF Feature	Manual?
1	Stakeholder Expectations Definition		
2	Technical Requirements Definition		
3	Logical Decomposition		
4	Design Solution Definition		
5	Product Implementation		
6	Product Integration		
7	Product Verification		
8	Product Validation		
9	Product Transition		
10	Technical Planning		
11	Requirements Management		
12	Interface Management		
13	Technical Risk Management		
14	Configuration Management		
15	Technical Data Management		
16	Technical Assessment		
17	Decision Analysis		

Discussion question: Which processes are well-supported by the tool? Which ones require the team's judgment and cannot be automated?

Part B: Lifecycle Phase Identification

For each activity below, identify which NASA lifecycle phase it belongs to and what review gate follows:

Activity	NASA Phase	Review Gate
“We need to monitor sea ice thickness”		
Writing system-level requirements		
Selecting a reaction wheel vendor		
Running thermal-vacuum tests		
Operating the satellite in orbit		
Decommissioning the satellite		
Preliminary orbit trade study		
Detailed structural analysis (FEA)		

Part C: CDF Position Interaction Map

Draw lines between positions that must interact to resolve each design issue:

Issue 1: The payload needs 50W but the solar array can only provide 30W. - Positions involved: _____

Issue 2: The reaction wheel vibration exceeds the payload stability requirement. - Positions involved: _____

Issue 3: The selected orbit requires a propulsive deorbit but there's no mass budget for propulsion. - Positions involved: _____

Issue 4: The X-band downlink interferes with the payload's sensitive receiver. - Positions involved: _____

Part D: SpaceCDF Exploration

Navigate through the SpaceCDF tool and answer:

1. How many KPI cards are shown on the Dashboard? List them:

1. What are the MCR exit criteria in the Gate Review tab?

1. What positions are available when starting a session?

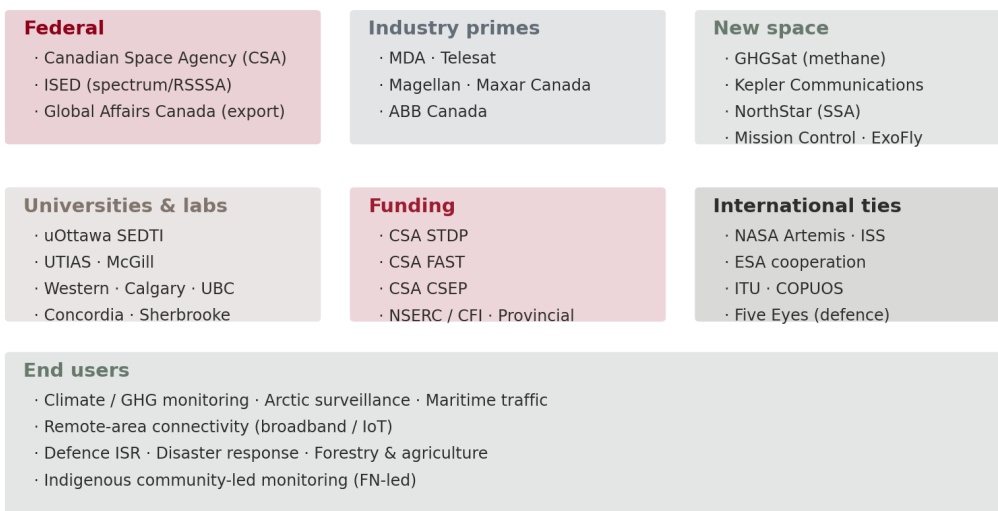
1. What mission types can be selected in the Requirements step?

Notes & Discussion Points

Use this space to record key points from group discussion:

Worksheet 1.2: Mission Need & Stakeholder Analysis

The Canadian Space Sector — illustrative map of actors



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Reference: Canadian space sector map.

Name: _____ Date: _____

Part A: Stakeholder Matrix

Identify at least 4 stakeholders for your team's mission. For each, capture their needs, constraints, and priority.

#	Stakeholder	Role	Key Needs	Constraints They Impose	Priority (P/S)
1					
2					
3					
4					
5					

Discussion: Which stakeholder has the most influence? Which has needs that conflict with others?

Part B: Objective Quality Checklist

Write 2-3 mission objectives below. Then score each against the checklist.

Objective 1: _____

Criterion	Yes/No	Notes
States WHAT, not HOW?		
Has a measurable criterion with number + unit?		
Is achievable with current technology?		
Is relevant to the problem statement?		
Is traceable to a stakeholder need?		
Is classified as primary or secondary?		

Objective 2: _____

Criterion	Yes/No	Notes
States WHAT, not HOW?		
Has a measurable criterion with number + unit?		
Is achievable with current technology?		
Is relevant to the problem statement?		
Is traceable to a stakeholder need?		
Is classified as primary or secondary?		

Part C: Problem Statement Writing

Write your team's mission problem statement below. It should answer: - What is the problem (capability gap)? - Who is affected? - What is the impact of not solving it? - What constraints exist?

Problem Statement:

Peer review: Exchange with another team. Does their problem statement prescribe a solution (HOW)? Mark any solution language.

Part D: MoE -> MoP -> TPM Chain

For one of your objectives, trace the measurement hierarchy:

Objective: _____

MoE (operational effectiveness from user perspective):

MoP (technical performance measure):

TPM (design parameter to track during development):

Part E: SpaceCDF Tool Entry

In SpaceCDF Step 1 (Mission Need), enter your problem statement, stakeholders, and objectives. Then answer:

1. Did the tool auto-detect your mission type from the objectives? What type did it suggest?

1. How many stakeholders did you enter? Are there any the tool doesn't let you capture?

1. What measurable criteria did you enter for each objective?

Notes

Worksheet 2.1: Requirements Engineering

Name: _____ Date: _____

Part A: Requirement Quality Assessment

Evaluate each requirement against the SMART checklist. Mark Pass (P) or Fail (F). Rewrite any failures.

#	Requirement	S	M	A	R	T	HOW?	Re-write
1	"The spacecraft shall operate at 500 km altitude"							
2	"The system shall achieve GSD <= 10 m at nadir"							
3	"The EPS shall use triple-junction GaAs cells"							
4	"The system shall have a design lifetime of at least 3 years"							
5	"The structure shall be made of aluminium 7075"							

Part B: Requirement Hierarchy

For your mission, write one requirement at each level:

Mission Requirement: _____

System Requirement (derived): _____

Subsystem Requirement (derived): _____

Draw the traceability: which objective does the mission requirement trace to? Which function does the subsystem requirement derive from?

Part C: Splitting Compound Requirements

Split this compound requirement into individually testable statements:

"The system shall provide 10m GSD multispectral imagery with 5-day revisit, 24-hour data latency, and positive power margin in all operational modes including eclipse."

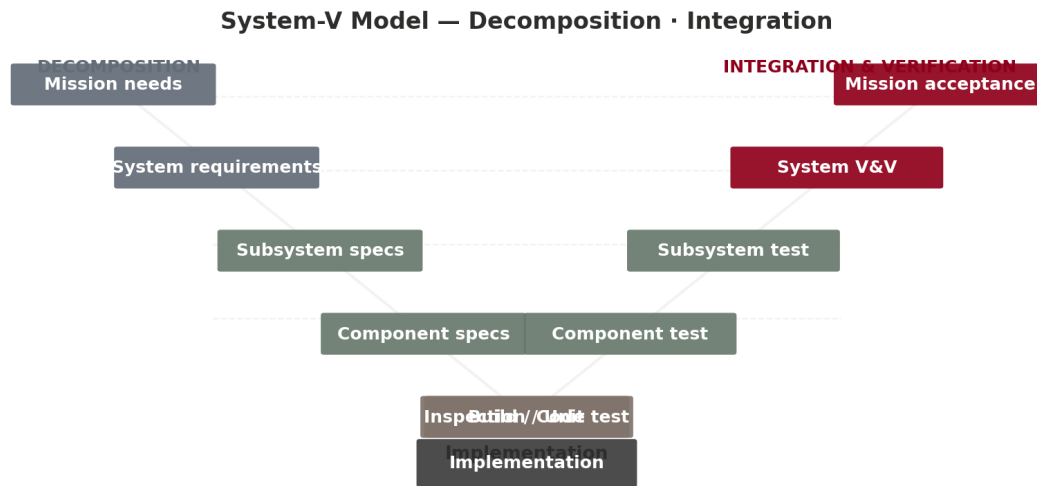
#	Split Requirement	Verification Method (A/T/R/I)
1		
2		
3		
4		
5		

Part D: SpaceCDF Requirements Generation

1. In SpaceCDF, click “Generate from Objectives.” How many requirements were generated? ____
2. Use the level filter – how many at each level? - Mission: _ **System:** _ Subsystem: ____
3. Find one requirement that says HOW not WHAT. Write it and your correction: - Original: _____ - **Corrected:** _____
4. Accept at least 5 requirements and reject at least 1. Record your decisions:

Req ID	Action	Rationale

Worksheet 2.2: Functional Decomposition



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Reference: System-V model.

Name: _____ Date: _____

Part A: Function Tree

Draw your mission's function tree (minimum 3 levels). Use the format:

```

F-001: [Primary mission function]
  ??? F-002: [Subfunction 1]
  ??? F-003: [Subfunction 2]
  ??? F-004: [Subfunction 3]
F-005: [Universal function 1]
F-006: [Universal function 2]
...
  
```

Your function tree:

Part B: Function-to-Requirement Traceability

For each leaf function, write one derived requirement:

Function ID	Function Name	Allocated To	Derived Requirement

Part C: Multi-Allocation Analysis

Identify at least one function that could be allocated to more than one subsystem:

Function: _____

Option A allocation: _____

Option B allocation: _____

Interface created: _____

Recommended boundary: _____

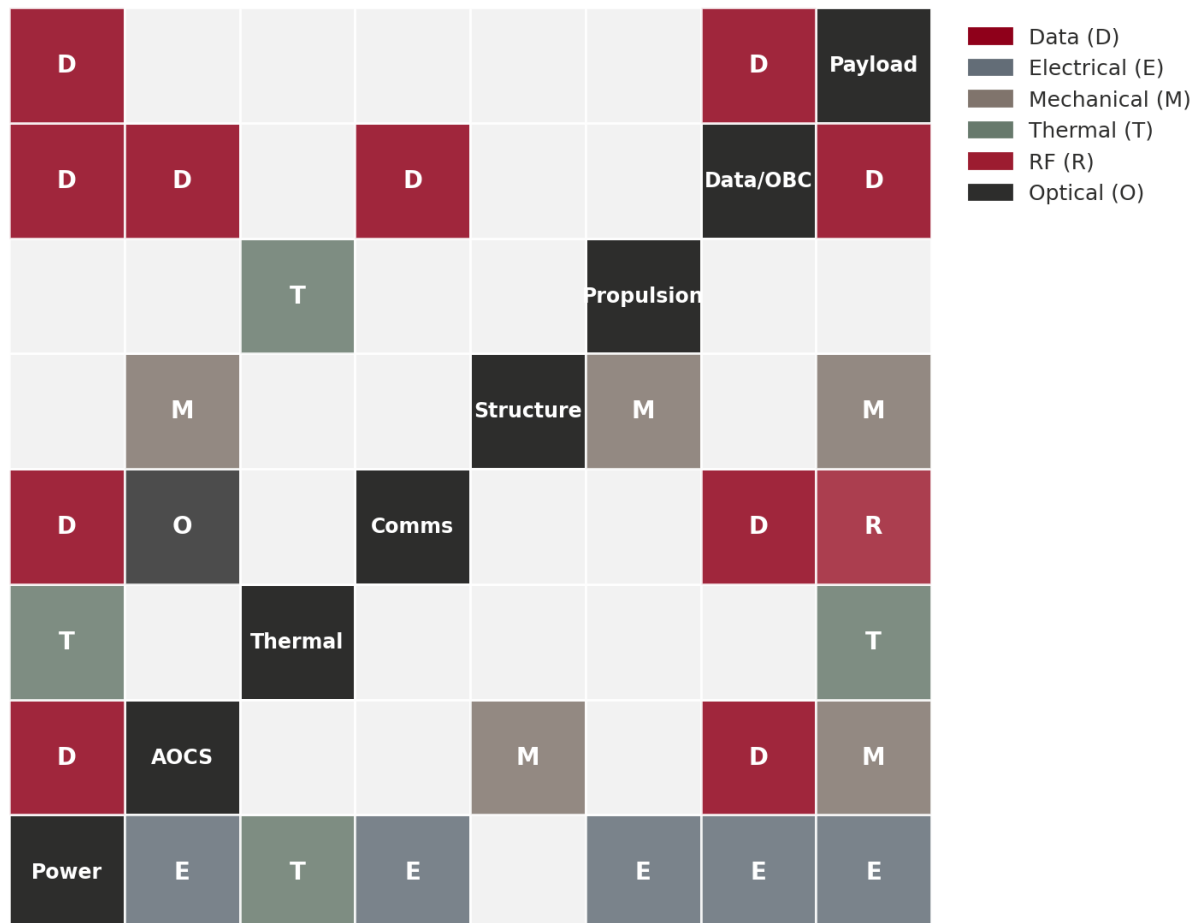
Part D: Coverage Check

After entering functions in SpaceCDF, check for coverage gaps:

- How many leaf functions have NO linked requirements? _____
- List them: _____
- What requirements would you derive for these?

Worksheet 2.3: Interface Management

N² Interface Matrix — 6U CubeSat (illustrative)



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Reference: an example N^2 interface matrix.

Name: _____ Date: _____

Part A: Interface Identification

List the 5 most critical interfaces in your design:

#	Subsystem A	Subsystem B	Interface Types	Key Concern
1				
2				
3				
4				
5				

Part B: Interface Requirements

Write 2 formal interface requirements for your most critical interface pair:

Interface: _ <-> _

IR-001: _____

IR-002: _____

Are these verifiable? By what method (A/T/R/I)? ____

Part C: Conflict Resolution

From the SpaceCDF Interface Matrix, identify one conflict (red border):

Conflict: _____

Severity: Critical / Major / Minor

Resolution options considered: 1. _____ 2. _____ 3. _____

Selected resolution: _____

Rationale: _____

Part D: Discussion

Which interface do you expect will cause the most problems during integration? Why?

Worksheet 2.4: Design Budgets

Name: _____ Date: _____

Part A: Mass Budget

Complete from your SpaceCDF design:

Subsystem	CBE Mass (kg)	% of Dry	Equipment Margin (20%)	MEV (kg)
Payload				
EPS				
AOCS				
TTC				
OBC				
Thermal				
Structure				
Harness				
Propulsion				
Dry Total		100%		
System margin (20%)				
Dry MEV				
Propellant				
Wet Mass				
Allocation				
Margin				

Part B: Power Budget by Mode

Subsystem	Safe (W)	Idle (W)	Imaging (W)	Downlink (W)	Eclipse (W)
OBC					
AOCS					
Payload					
TTC					
Thermal					
Total					
Duty cycle %					

Orbit-average power: $P_{avg} = \text{_____ W}$

SA power needed: $P_{SA} = P_{peak} + \text{recharge} = \text{__} + \text{__ W}$

SA BOL (with degradation): _____ W

Part C: Budget Health Check

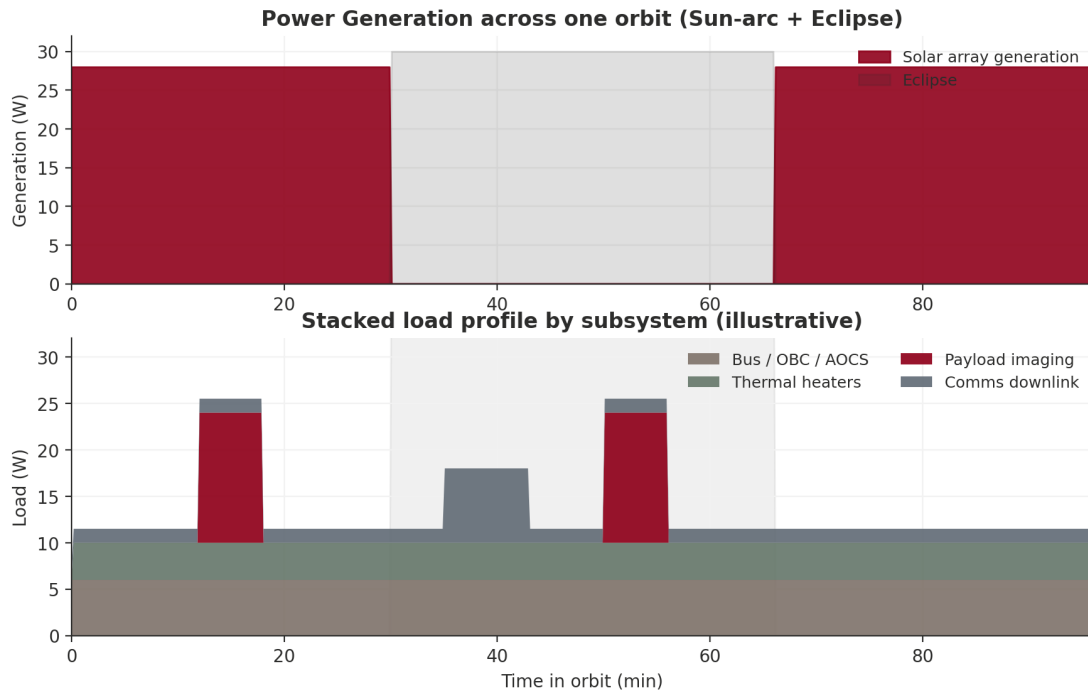
From the SpaceCDF Dashboard, record:

Budget	Margin	Status (G/A/R)
Mass	_____ %	
Power	_____ %	
Link	_____ dB	
Cost	__ MEUR vs __ ceiling	
Pointing	__° vs __° req	

Which budget is tightest? _____

What design change would improve it? _____

Worksheet 3.1: Orbit Design & Selection



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Reference: power profile across one orbit.

Name: _____ Date: _____

Part A: Orbital Mechanics Calculations

Your selected orbit: Altitude $h = \underline{\hspace{1cm}}$ km, Inclination $i = \underline{\hspace{1cm}}^\circ$, Type: $\underline{\hspace{1cm}}$

Compute (show working):

1. **Semi-major axis:** $a = R_E + h = 6371 + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ km = $\underline{\hspace{1cm}}$ m
2. **Orbital period:** $T = 2\pi\sqrt{a^3/\mu} = \underline{\hspace{1cm}}$ s = $\underline{\hspace{1cm}}$ min
3. **Orbital velocity:** $v = \sqrt{\mu/a} = \underline{\hspace{1cm}}$ m/s
4. **Eclipse fraction:** $f = (1/\pi) \times \arccos(\sqrt{1 - (R_E/a)^2}) = \underline{\hspace{1cm}}$
5. **Sunlight time per orbit:** $t_{\text{sun}} = T \times (1-f) = \underline{\hspace{1cm}}$ min
6. **Eclipse time per orbit:** $t_{\text{ecl}} = T \times f = \underline{\hspace{1cm}}$ min

Part B: Orbit Trade Matrix

From the SpaceCDF orbit trade advisor, record the top 3 candidates:

Rank	Orbit	Alt (km)	Inc (°)	GSD (m)	Revisit (d)	Lifetime (yr)	5yr?	Score
1								
2								
3								

Selected orbit: _ Rationale: _____

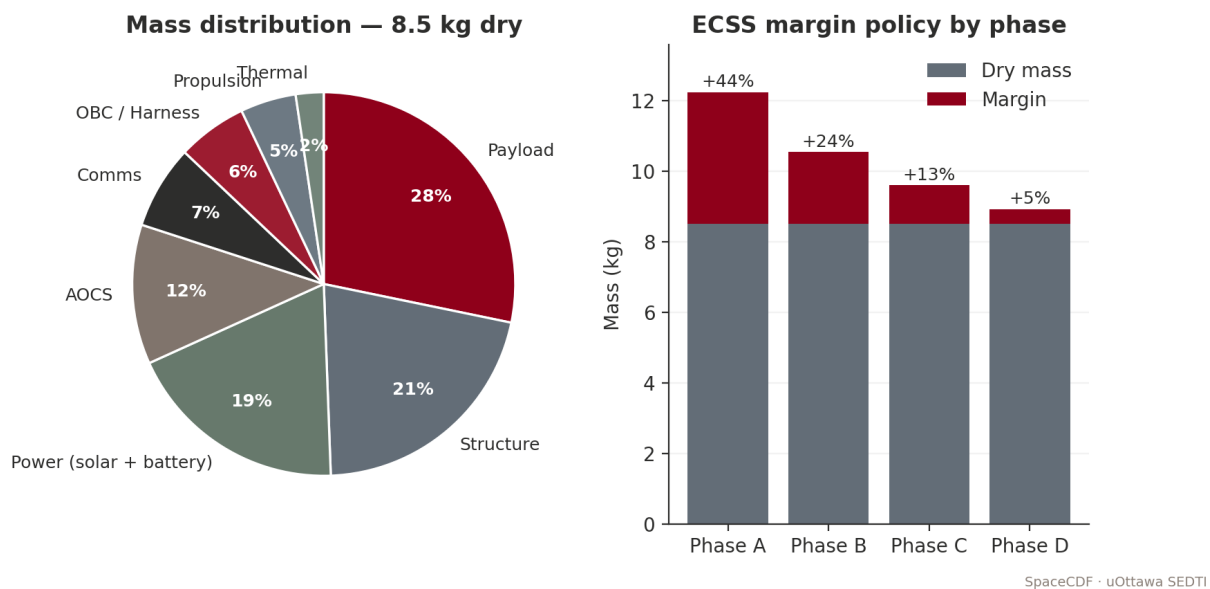
Part C: Debris Compliance

For your selected orbit: - Natural lifetime: _ years - FCC 5-year compliant? Y / N - IADC 25-year compliant? Y / N - Propulsion needed for deorbit? Y / N - If yes, estimated deorbit ?V: _ m/s

Part D: Discussion

What would happen to your mission if you moved 100 km higher? 100 km lower?

Worksheet 3.2: Payload & Communications



Reference: mass distribution and ECSS margin.

Name: _____ Date: _____

Part A: Payload Sizing (Optical missions – skip if non-optical)

GSD requirement: _ m Altitude: _ km Wavelength: 0.55 μm

1. Diffraction-limited GSD: $\text{GSD}_{\text{diff}} = 1.22 \times \lambda \times h / D = _ \text{ m}$ (for $D = _ \text{ m}$)
2. Required aperture for target GSD: $D \geq 1.22 \times \lambda \times h / \text{GSD} = _ \text{ m}$
3. Payload mass estimate: $M = 20 \times D^{1.5} + 2 = _ \text{ kg}$
4. Payload power estimate: $P = 3 \times M = _ \text{ W}$

Part B: Link Budget

Complete the link budget for your mission's primary downlink:

#	Parameter	Value	Unit
1	TX Power		dBW
2	TX Antenna Gain		dBi
3	TX Losses		dB
4	EIRP		dBW
5	Free Space Path Loss		dB
6	Atmospheric Loss		dB
7	Pointing Loss		dB
8	RX Antenna Gain		dBi
9	System G/T		dB/K
10	C/N0		dBHz
11	Data Rate (10log10)		dBbps
12	Eb/N0 available		dB
13	Eb/N0 required		dB
14	Implementation margin		dB
15	LINK MARGIN		dB

Does the link close (≥ 3 dB)? Y / N

Part C: Frequency Band Selection

License type chosen: Amateur / Experimental / Commercial

Band selected: _ Frequency: ____ MHz

Rationale: _____

Filing required: _____

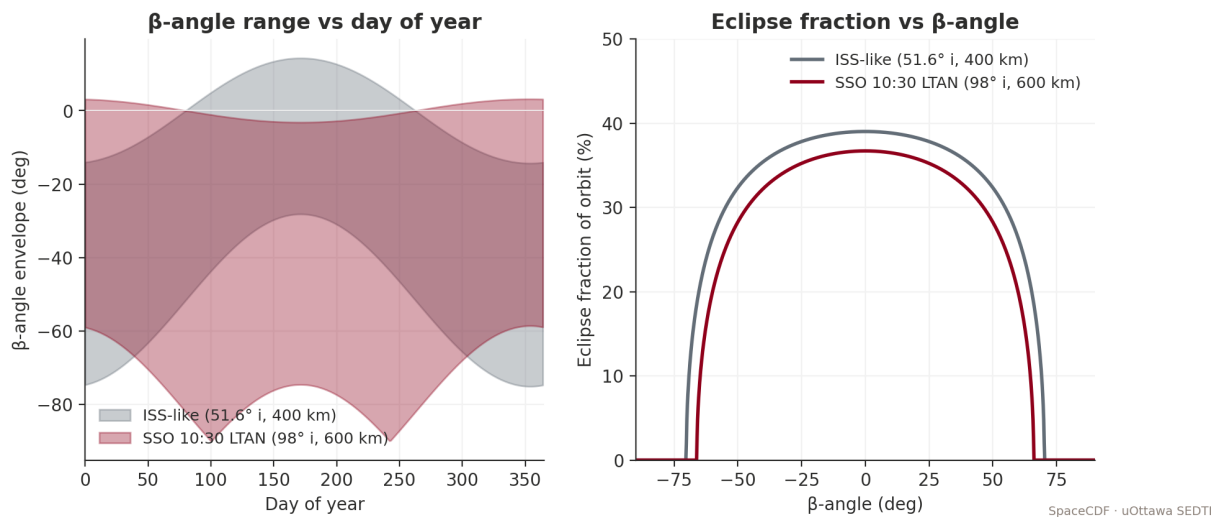
Part D: SpaceCDF Link Budget Tool

Open the Link Budget tab and enter your parameters. Compare to your hand calculation:

Parameter	Hand Calc	SpaceCDF	Difference
EIRP			
FSPL			
Link Margin			

Worksheet 3.3: Power, AOCS, & Thermal

β -angle behaviour over a year — and eclipse fraction analytics



Reference: β -angle and eclipse fraction.

Name: _____ Date: _____

Part A: Solar Array Sizing (show calculations)

1. Peak sunlight power demand: $P_{\text{peak}} = _ \text{ W (from mode: _)}$
2. Eclipse power demand: $P_{\text{eclipse}} = _ \text{ W}$
3. Eclipse time: $t_{\text{ecl}} = _ \text{ min; Sunlight time: } t_{\text{sun}} = _ \text{ min}$
4. Recharge power: $P_{\text{rech}} = (P_{\text{ecl}} \times t_{\text{ecl}}) / (t_{\text{sun}} \times 0.9) = _ \text{ W}$
5. **SA EOL required:** $P_{\text{SA}} = P_{\text{peak}} + P_{\text{rech}} = _ \text{ W}$
6. EOL degradation factor (3yr, 2.5%/yr): $(1 - 0.025)^3 = _$
7. **SA BOL required:** $P_{\text{SA_BOL}} = P_{\text{SA_EOL}} / \text{factor} = _ \text{ W}$
8. SA area: $A = P_{\text{BOL}} / (0.295 \times 1361 \times 0.85) = _ \text{ m}^2$
9. SA type needed: Body-mounted / Single deploy / Dual deploy

Part B: Battery Sizing

1. Eclipse energy: $E = P_{\text{ecl}} \times t_{\text{ecl}} / 60 = _ \text{ Wh}$
2. Battery capacity: $C = E / (\text{DoD} \times \eta) = _ / (0.3 \times 0.95) = _ \text{ Wh}$
3. Battery mass: $m = C / 150 = _ \text{ kg}$

Part C: AOCS Selection

Pointing requirement: _____ °

AOCS architecture selected: (tick one) - ☐ Passive magnetic (>5°) - ☐ Magnetorquers only (2-5°) - ☐ Reaction wheels + MTQ (0.1-2°) - ☐ Fine pointing: RW + star tracker + MTQ (<0.1°)

Part D: Pointing Error Budget

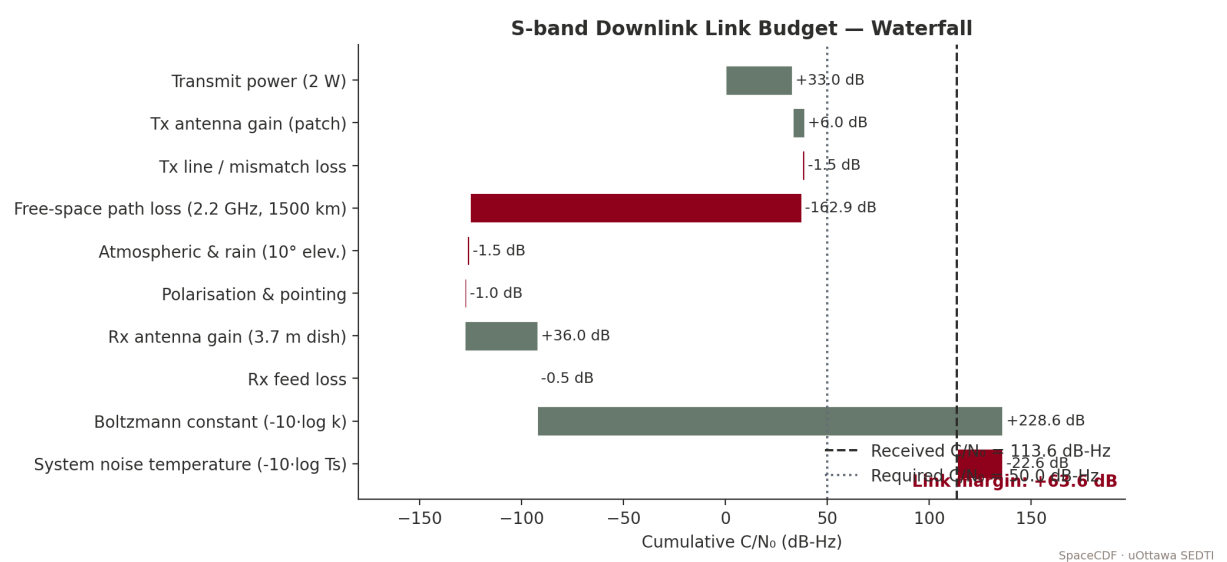
Error Source	Value (°)	Value^2
Sensor accuracy		
Actuator resolution		
Alignment knowledge		
Thermal distortion		
Jitter		
Orbit knowledge		
RSS Total	=?(?) =	

Requirement: _ ° Margin: ° (__ %)

Part E: Thermal Check

From SpaceCDF Dashboard, record: - Predicted hot case temp: _ °C - **Predicted cold case temp: °C** - **Component max operating: °C** -> **Hot margin: _ °C (need >=5°C)** - **Component min operating: °C** -> **Cold margin: °C (need >=5°C)**

Worksheet 3.4: Structure, Propulsion, & Equipment Selection



Reference: link-budget waterfall.

Name: _____ Date: _____

Part A: CDS Compliance Check

CDS Requirement	Your Design	Compliant?
Form factor (___U) dimensions		
Max mass (___kg)		
Rail material (anodised Al)		
Deployment switches (min 2)		
RBF pin provided		
No protrusions beyond rails		
CG within 2 cm of geometric centre		

Part B: Propulsion Decision

Does your mission need propulsion? - Orbit altitude: km - **Natural lifetime:** years - FCC 5-yr compliant without propulsion? Y / N - Orbit maintenance needed? Y / N - Constellation phasing needed? Y / N

Decision: Propulsion / No propulsion

If propulsion: Type: Isp: s ? **V needed:** m/s **Propellant mass:** $m = m_{dry} \times (e^{(V/(Isp \times 9.81))} - 1) =$ kg

Part C: Equipment Selection Log

Category	Component Selected	Manufacturer	Mass (kg)	Power (W)	Cost (kEUR)	Qty
	EPS Board					
	Battery					
	Solar Panels					
	OBC					
	Reaction Wheel					x
	Magnetorquer					x
	Star Tracker					
	Sun Sensor					x
	Transponder					
	Antenna					
	Structure					
	TOTAL					

Parametric estimate: kg **Selected total:** kg **Difference:** kg

Part D: Component Trade Study

Subsystem traded: **Components compared:** 3

Criterion	Weight	Option A: _	Option B: _	Option C: _
Mass	0.20			
Power	0.15			
Cost	0.20			
TRL	0.15			
Heritage	0.15			
Performance	0.15			
Weighted Score				

Winner: __ Rationale: _____

Worksheet 4.1: Equipment Selection & Integration

Technology Readiness Level (TRL) Ladder

TRL 9	Flight-proven	TRL 1-3 Discussed in Course Plan §2 — research / IRAD context
TRL 8	Qualified system	
TRL 7	Operational demo	
TRL 6	Relevant-environment demo	TRL 4-6 CubeSat platforms typically buy at TRL ≥ 6 (commercial COTS)
TRL 5	Relevant-environment validation	
TRL 4	Lab validation	TRL 7 First in-orbit demonstration — relevant for new instruments
TRL 3	Proof of concept	
TRL 2	Concept formulated	TRL 8-9 Operational missions — flight heritage drives selection in CDF Day 3
TRL 1	Basic principles	

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Reference: TRL ladder.

Name: _____ Date: _____

Part A: Compatibility Verification

For each selected component, verify interface compatibility:

Component	Bus Voltage OK?	Protocol Compatible?	RF Band Match?	Notes

Part B: Budget Tracking (from Equipment Browser)

After all selections, record the live budget bar values:

- Total selected mass: _____ kg
- Total selected power: _____ W
- Total selected cost: _____ kEUR
- Mass allocation: _ kg -> Margin: kg (__ %)
- Volume utilisation: _ % of _U form factor

Does everything fit? Y / N – If not, what would you change?

Part C: Incompatibility Found and Resolved

Describe one incompatibility detected during selection and how you resolved it:

Issue: _____

Components involved: _____

Resolution: _____

Worksheet 4.2: Verification & Validation Planning

Name: _____ Date: _____

Part A: V&V Method Assignment

For 10 key requirements, assign method (A/T/R/I), phase, and level:

#	Requirement ID	Requirement Text (abbreviated)	Method	Phase	Level	Justification
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Part B: Dual-Method Requirements

Identify 3 requirements that need BOTH analysis AND test:

Requirement	Analysis (Phase B)	Test (Phase C/D)

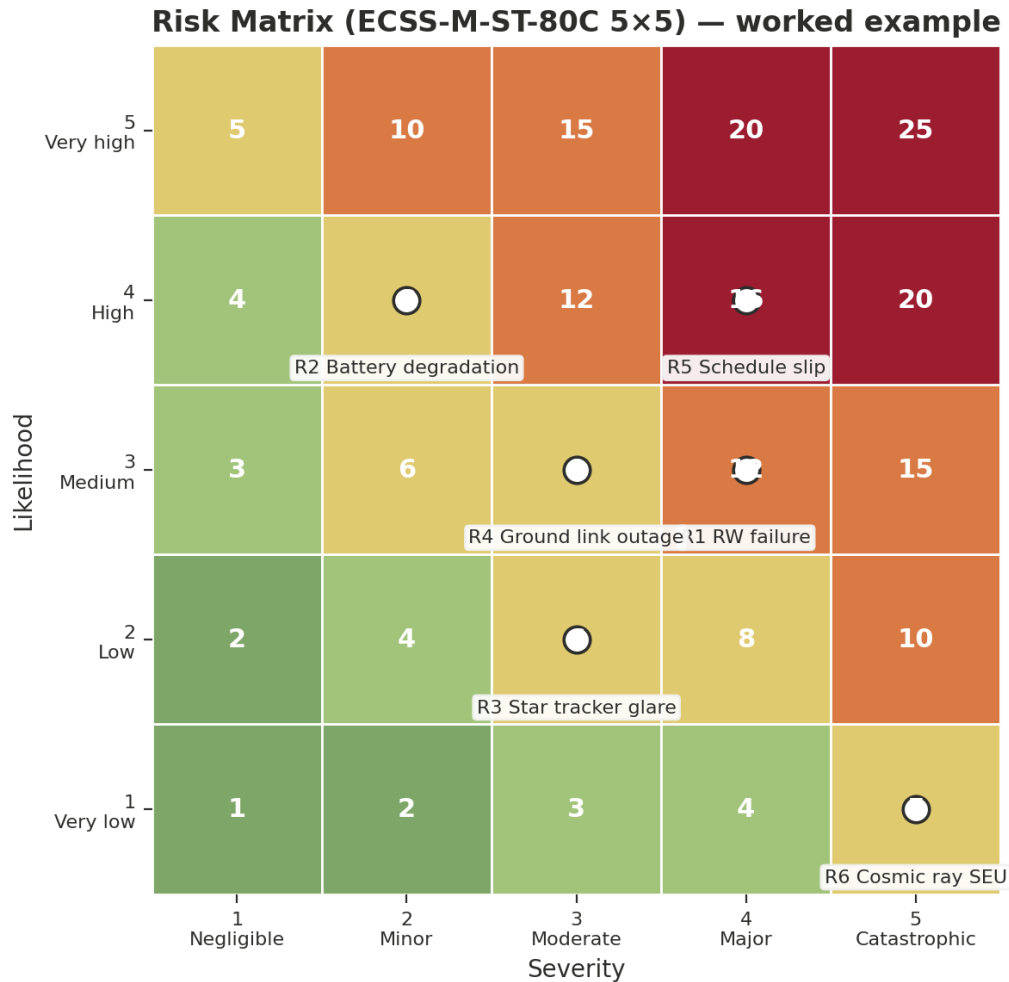
Part C: Environmental Test Sequence

For your selected launch vehicle, specify the test sequence:

1. ☐ Initial functional test
2. ☐ ___ vibration (levels: **_ gRMS**, axes: **3**, duration: **_**/axis)
3. ☐ Post-vibe functional test
4. ☐ Thermal vacuum (hot: **+_°C**, cold: **-°C**, cycles: **_**)
5. ☐ Post-TVAC functional test
6. ☐ EMC test (required? Y/N)
7. ☐ Mass measurement
8. ☐ Deployer fit check

Launch vehicle: ___ **PUG used:** ___

Worksheet 4.3: Risk Management



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Reference: 5×5 risk matrix.

Name: _____ Date: _____

Part A: Risk Register

#	Risk Description	L (1-5)	C (1-5)	Score	Category	Mitigation Strategy	Owner
1							
2							
3							
4							
5							

Scoring: 1-4 Low (accept), 5-9 Medium (mitigate if practical), 10-15 High (must mitigate), 16-25 Critical (redesign)

Part B: Single-Point Failure Analysis

Component	Failure Mode	Effect on Mission	SPF?	Acceptable?	Mitigation

Part C: Discussion

What is the highest-risk item in your design? What would it take to bring the risk to an acceptable level?

Worksheet 4.4: Cost Estimation

Name: _____ Date: _____

Part A: Mission Cost Estimate

WBS Element	Parametric Est. (kEUR)	Bottom-Up Est. (kEUR)	Notes
Programme Management			
Systems Engineering			
Mission Assurance			
Payload			
Bus Hardware			
Integration & Test			
Software (FSW + GSW)			
Launch			
Ground Segment			
Operations (___ years)			
TOTAL			

P50 estimate: _ kEUR P80 estimate (×1.3): ___ kEUR

Part B: Constellation Cost (if applicable)

Number of satellites: _ Learning rate: __%

Unit #	Unit Cost (kEUR)	Cumulative (kEUR)
1		
2		
4		
8		
Total (N=___)		

- Spares (15%): **_ units** × **_ kEUR** = **_____ kEUR**
- Launch: **_ sats** × **_ kEUR** = **_____ kEUR**
- Ground + Ops: **_____ kEUR**

Total constellation cost: **_ kEUR** = **___ MEUR**

Part C: Cost Drivers

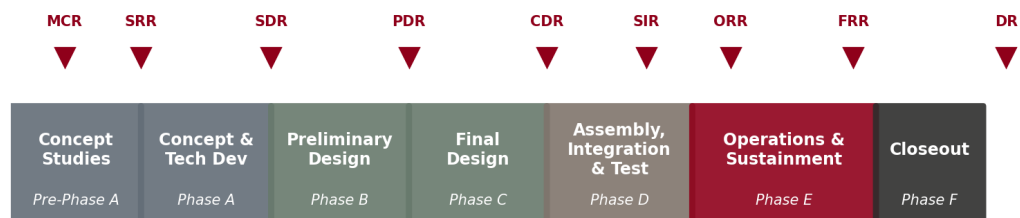
Top 3 cost drivers in your mission: 1. _____ 2. _____ 3. _____

What would you change to reduce cost by 20%? _____

Worksheet 5.1: Gate Review Preparation

Project Lifecycle Phases & Review Gates — NASA vs ECSS

NASA gates



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Reference: gate sequence.

Name: _____ Date: _____

Part A: MCR Exit Criteria Status

From SpaceCDF Gate Review tab, record each criterion:

#	Criterion	Status (Pass/Fail/Manual)	Evidence	If Fail: Fix Action
EC-01	Mission need defined			
EC-02	Stakeholders identified			
EC-03	Objectives with criteria			
EC-04	Alternatives considered			
EC-05	Concept justified			
EC-06	ConOps documented			
EC-07	Feasible concept			
EC-08	Sustainable (debris)			
EC-09	Requirements traceable			
EC-10	Interfaces managed			

Overall readiness: READY / NOT READY **Must-pass met:** /

Part B: MCR Presentation Outline

Prepare your team's 12-minute MCR presentation:

1. Mission Need (2 min): _____
 2. Why Space? (2 min): _____
 3. System Design (3 min): _____
 4. Feasibility (2 min): _____
 5. Risks (2 min): _____
 6. Actions (1 min): _____
-

Part C: Action Items from Review

#	Action	Responsible	Deadline	Acceptance Criteria
1				
2				
3				

Board decision: GO / NO GO / GO with actions

Worksheet 5.2: Regulatory & Licensing

Name: _____ Date: _____

Part A: Filing Requirements Checklist

Tick all that apply to your mission:

- ☐ ISED spectrum licence (CPC-2-6-02) – all missions transmitting
- ☐ ITU API/coordination/notification – commercial/experimental
- ☐ IARU amateur coordination – if using amateur bands
- ☐ RSSSA operating licence – if ANY Earth imaging capability
- ☐ Export permit – if controlled components or US launch
- ☐ Controlled Goods registration – if handling US-origin controlled items
- ☐ COPUOS registration – all missions (post-launch)
- ☐ FCC Part 25 – if US-licensed or US launch
- ☐ End-of-life compliance report – all missions

Part B: Licensing Decision

Your license type: Amateur / Experimental / Commercial

Frequency band: _____ Data rate: _____ Mbps

ITU filing needed? Y / N Estimated timeline: _____ months

RSSSA needed? Y / N Reason: _____

Export control concerns? Y / N Components of concern: _____

Part C: Regulatory Timeline

Working backward from your target launch date:

Milestone	Date (L minus)	Filing	Authority
L - __ months		ITU API	ISED->ITU
L - __ months		ISED spectrum	ISED
L - __ months		RSSSA (if needed)	Global Affairs
L - __ months		Export permits	Global Affairs
L - __ months		IARU coord (if amateur)	RAC->IARU
L + 1 month		COPUOS registration	Global Affairs

Critical path filing: _____

Part D: SpaceCDF Templates

Generate these from the Exports tab and note TBD fields:

Document	# Auto-filled fields	# TBD fields	Key TBDs
ITU API			
RSSSA			
Export Assessment			
COPUOS Registration			
End-of-Life Report			

Worksheet 5.3: Launch Integration

Name: _____ Date: _____

Part A: Launch Provider Selection

From SpaceCDF Launch Selector, record top 3 options:

Provider	Vehicle	Price	Capacity	Your Mass	Fits?	Deployer
----------	---------	-------	----------	-----------	-------	----------

Selected: ____ Rationale: _____

Mass allocation from provider: _ kg -> Mass margin: kg (__ %)

Part B: ICD Compliance Checklist

ICD Requirement	Your Design	Compliant?	Evidence
Dimensions within deployer envelope			
Mass within deployer limit			
CG within 2 cm of geometric centre			
2+ deployment switches on rails			
RBF pin provided			
No protrusions beyond rails (stowed)			
Battery <= 50% SoC at delivery			
No RF emissions in deployer			
Antennas stowed clear of deployer			

Part C: Environmental Test Specification

From launch vehicle PUG, specify your test levels:

Random vibration: _ gRMS, min/axis, 3 axes **Sine vibration:** g, _ - Hz **Shock:** g at _ Hz
TVAC hot: predicted max + 10°C = °C **TVAC cold:** predicted min - 10°C = __ °C **TVAC**
cycles: ____

Part D: Discussion

What ITAR implications arise from launching your Canadian CubeSat on a US vehicle?

Worksheet 5.4: Optimisation & Final Design

Name: _____ Date: _____

Part A: Sensitivity Analysis

From SpaceCDF Optimizer (or Trade Studies), identify which parameters most influence your design:

Variable	mu* (importance)	sigma (non-linearity)	Classification
Key driver / Negligible / Investigate			

Most important parameter: _____

Action: _____

Part B: Final Design Summary

Parameter	Value	Unit	Margin	Status
Dry mass		kg	___%	G/A/R
Wet mass		kg		
SA power EOL		W	___%	G/A/R
Battery capacity		Wh		
Link margin		dB		G/A/R
Pointing accuracy		°	___%	G/A/R
Total cost		MEUR		
Data balance		GB/day surplus		G/A/R
Debris compliance		/100		G/A/R

Part C: Documents Generated

Tick each document generated from SpaceCDF:

- ☐ MRD (Mission Requirements Document)
 - ☐ Technical Specification
 - ☐ Verification Plan
 - ☐ ConOps Document
 - ☐ SEMP
 - ☐ Risk Management Plan
 - ☐ IRD
 - ☐ Test Plan
 - ☐ BOM
 - ☐ ITU Filing Template
 - ☐ RSSSA Filing
 - ☐ End-of-Life Report
 - ☐ COPUOS Registration
-

Part D: Lessons Learned

1. What was the most challenging design decision?

1. What would you do differently if starting over?

1. What does the design need next (Phase B activities)?

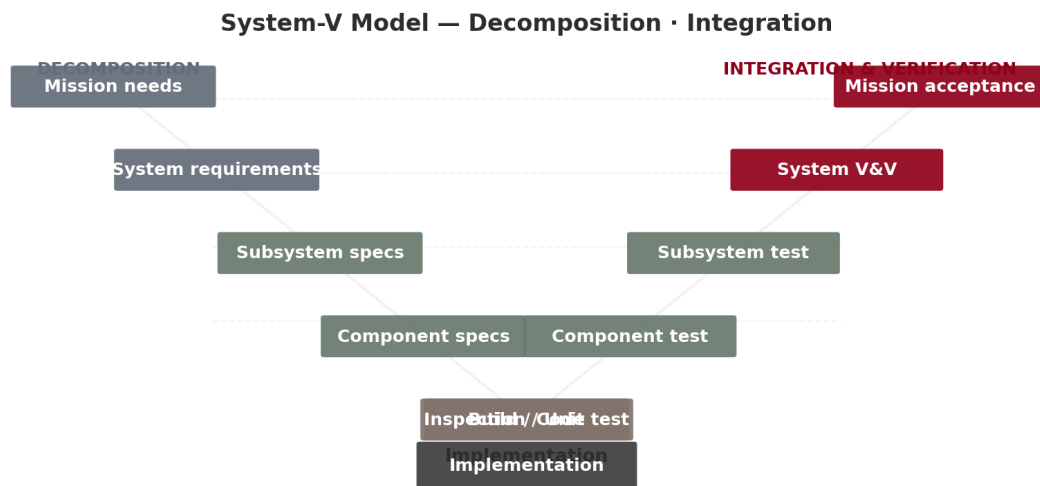
1. What was the most valuable feature of the tool?

1. What feature was missing that you needed?

Final Presentation Peer Review (completed by reviewing team)

Criterion	Score (/10)	Comments
Problem clearly defined		
Design feasibility (budgets close)		
Trade study rigour		
Requirements quality		
Completeness		
Communication clarity		
Risk awareness		
TOTAL	/70	

Worksheet: System Architecture Selection



SpaceCDF · uOttawa SEDTI

Reference: System-V model — left side.

Name: _____ Date: _____

Part A: Architecture Decisions

For each subsystem, record your selected architecture, the key rationale, and the primary derived requirement:

Subsys-tem	Options Con-sidered	Selected Architec-ture	Ra-tionale	Key Derived Require-ment
EPS				
AOCS				
TTC				
Thermal				
Structure				
Propulsion				
OBC				
Ground				

Part B: Architecture Trade Study

For one contested subsystem (where 2+ options were viable), complete a formal trade:

Subsystem: ____

Criterion	Weight	Option A: _	Option B: _	Option C: _
Mass	0.20			
Power	0.15			
Cost	0.20			
TRL / Heritage	0.20			
Performance	0.15			
Risk	0.10			

Weighted Score

Winner: ____ Rationale: _____

Part C: Derived Requirements Review

From SpaceCDF Architecture tab, list all derived requirements for your AOCS selection:

Req ID	Level	Requirement Text	Verification Method

Are any requirements missing? What would you add?

Part D: Block Diagram Analysis

For your selected TTC architecture, draw the block diagram showing: - All components (blocks) - Signal flow (connections with labels) - Interface to other subsystems (OBC, AOCS for antenna pointing)

Part E: Discussion

1. Which architecture decision had the biggest impact on your overall design?

1. Where did architecture choices create new interface requirements?

1. Did any architecture selection conflict with another subsystem's choice?

Reference list

A live, hyperlinked list of standards and references used across the worksheets.

- **NASA SEH** — <https://www.nasa.gov/reference/systems-engineering-handbook/>
- **NASA CubeSat 101** — https://www.nasa.gov/wp-content/uploads/2017/03/nasa_c-sli_cubesat_101_508.pdf
- **ECSS standards** — <https://ecss.nl/>
- **Cal Poly CDS Rev 14** — <https://www.cubesat.org/cds-announcement>
- **CCSDS PUS** — <https://public.ccsds.org/Pubs/660x0g3.pdf>
- **ITU Radio Regulations** — <https://www.itu.int/pub/R-REG-RR>
- **ISED CPC-2-6-02** — <https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en>
- **Remote Sensing Space Systems Act** — <https://laws-lois.justice.gc.ca/eng/acts/R-5.4/>
- **SMAD4** (Larson & Wertz) — Space Mission Analysis and Design, 4th ed.
- **SME** (Wertz, Everett & Puschell) — Space Mission Engineering (The New SMAD), 2011.